

Notes: Hong and Kubik (JF, 2003)

Analyzing the Analysts: Career Concerns and Biased Earnings Forecasts

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1 Big picture

- **Question.** What does the labor market for sell-side analysts actually reward: forecasting skill, stock-promotion ability, or both?
- **Motivation.** By the late 1990s, analysts were accused of keeping unrealistically optimistic views on firms—especially firms tied to investment banking business. The paper asks whether those accusations show up in the *career incentives* analysts face.
- **Core idea.** The authors do not observe compensation directly. Instead, they infer implicit incentives from **career outcomes**: whether analysts remain at, move up to, or move down from high-status brokerage houses.
- **Setting.** The paper uses a large panel of U.S. analysts from I/B/E/S between 1983 and 2000, covering roughly 12,336 analysts, 619 brokerage houses, and 8,441 firms.
- **Why job separations are informative.** There is a clear prestige hierarchy among brokerage houses. Working for Goldman Sachs or Merrill Lynch is not the same as working for a small regional specialist. Moving up that hierarchy is a favorable career outcome; moving down is unfavorable.
- **Main hypotheses.**
 - More accurate forecasters should be more likely to experience favorable job separations.
 - Controlling for accuracy, analysts who are relatively optimistic should also be more likely to experience favorable job separations if brokerage houses value stock promotion.
 - These patterns should be stronger for analysts covering stocks underwritten by their own houses.
 - The late-1990s market boom should weaken the reward to accuracy and strengthen the reward to optimism.
- **Main empirical strategy.** Construct analyst-level measures of relative forecast accuracy and relative forecast optimism over rolling three-year windows, then relate those lagged measures to future job transitions using probit models with brokerage-house, year, experience, coverage, and portfolio-composition controls.
- **Main result on accuracy.** Accurate analysts are rewarded. Being in the **bottom 10%** of relative accuracy raises the probability of moving *down* the hierarchy by about **4.8 percentage points** and lowers the probability of moving *up* the hierarchy by about **1.4 percentage points**. Being in the **top 10%** of relative accuracy raises the chance of moving up by about **1.1 percentage points**.
- **Main result on optimism.** Controlling for accuracy, relatively optimistic analysts are also rewarded. Being in the **top 10%** of relative optimism lowers the chance of moving down and raises the chance of moving up the hierarchy by almost **90%** relative to the baseline move-up rate.
- **Conflict-of-interest result.** For analysts covering stocks underwritten by their own houses, job separations become *less sensitive to accuracy* and *more sensitive to optimism*.
- **Time-variation result.** During 1996–2000, career outcomes are less tied to accuracy and somewhat more tied to optimism than in earlier years.

- **Internal-labor-market result.** Poorly performing analysts are more likely to be removed from high-profile stocks inside their brokerage houses, especially stocks with very high market capitalization or heavy analyst following.
- **Why the paper matters.** It moves the discussion away from pure anecdotes. The paper shows systematically that analysts are rewarded for expertise *and* for optimism, with the balance tilting more toward optimism in settings where conflicts of interest are strongest.
- **Economic takeaway.** The analyst labor market does not simply pay for information production. It also pays for behavior that helps brokerages attract underwriting business and trading commissions.

2 Data and measurement (key objects)

2.1 Main analyst sample

- **Data source.** I/B/E/S Detail Earnings Estimate History File.
- **Coverage period.** 1983–2000.
- **Scale of the panel.** The sample contains forecasts from about **12,336 analysts**, working for **619 brokerage houses**, covering **8,441 firms**.
- **Analyst specialization.** Analysts usually specialize by industry and follow, on average, about **9.3 firms per year** with a standard deviation of about **8.3**.
- **Experience measurement.** Because I/B/E/S starts in 1983, the authors can only measure analyst experience cleanly for analysts who begin after 1983. They therefore exclude **left-censored analysts** from the main tests.
- **Typical sample length.** Analysts remain in the I/B/E/S panel a bit over **four years on average**, with a standard deviation of about four years. The 90th percentile is **11 years**.

2.2 Brokerage-house structure and hierarchy

- The number of brokerage houses in the sample rises from about **90 in 1983** to more than **300 in 2000**.
- Average house size falls from about **20.7 analysts** in 1983 to about **11.6 analysts** in 2000, consistent with growth in smaller specialty research shops.
- The paper treats brokerage houses as lying on a prestige ladder:
 - **high-status houses:** traditional Wall Street powerhouses with prestige, compensation, and often investment-banking clout,
 - **low-status houses:** smaller, regional, or specialized brokerages.

2.3 Three measures of brokerage-house status

- **Primary measure: Institutional Investor (I.I.) ranking.** Each October, *Institutional Investor* lists “The Leaders,” the roughly top 10 houses with the most All-Americans. The paper

classifies those top 10 as **high-status** each year.

- **Alternative measure 1: house size.** The top 10 brokerage houses by number of analysts in a year are coded as high-status.
- **Alternative measure 2: Carter–Manaster ranking.** The top 10 houses by underwriter prestige are coded as high-status using the Carter–Manaster hierarchy.
- **Sanity check.** Using the I.I. ranking, high-status houses employ about **23%** of analysts in a typical year. Using the size ranking, the share is larger, roughly **25%–39%**. Using Carter–Manaster, it is roughly **18%–25%**.
- **Interpretation.** The exact ranking is somewhat arbitrary, but all three measures identify roughly the same set of elite firms, and the main results are robust across them.

2.4 Additional data sources

- **Underwriting relationships.** From the SDC New Issues database, the authors collect IPO dates and lead underwriters. A stock is treated as having an underwriting relationship with analyst i 's brokerage house if, within the year of the IPO or the following two years, the analyst forecasts that stock and her house was the lead underwriter.
- **All-American status.** From *Institutional Investor*, the authors identify First Team All-Americans each year. There are about **60 per year on average**.

2.5 Job-separation variables

- **Change house.** Analyst works for one house at the start of year t and moves to another house during year t .
- **Move up.** Analyst starts at a low-status house and moves to a high-status house.
- **Move down.** Analyst starts at a high-status house and moves to a low-status house.
- If a house itself changes status from one year to the next, the analyst is *not* counted as moving. Only actual separations count.

2.6 Descriptive mobility facts

- About **14.3%** of analysts change brokerage houses in a typical year.
- Among movers, about **12%** are upward moves, about **10%** are downward moves, and the rest are lateral.
- Among experienced analysts (those with at least three years of history), the unconditional probability of:
 - moving from low to high status is about **2.73%**,
 - moving from high to low status is about **7.77%**.
- **Interpretation.** Moving up is rare and competitive. Staying at the top is also hard.

2.7 Portfolio continuity across job changes

- The paper measures the fraction of an analyst’s portfolio in year $t + 1$ consisting of stocks not covered in year t .
- For the full sample, about **26.18%** of next year’s portfolio is new.
- Analysts who leave their houses have a new-stock share of about **27.59%**; analysts who stay have about **25.93%**.
- **Interpretation.** Movers and stayers look similar, which suggests brokerage houses partly hire analysts for the specific industry and firm knowledge they already developed.

2.8 Experience and funneling up the hierarchy

- The fraction of analysts working for high-status houses rises with experience.
- For example, among analysts observed for at least five years, about **19%** work for high-status houses in year 1, rising to about **23%** by year 5.
- In longer-lived cohorts, the fraction reaches roughly **25%–27%** by later years.
- **Interpretation.** Analysts are, on average, funneled upward over time. That is why experience controls are essential in the career regressions.

2.9 Observed versus unobserved

- **Observed.** Forecasts, realized earnings, stock coverage, brokerage affiliation, house size, job moves, underwriting links, and All-American status.
- **Not observed directly.** Pay, bonus contracts, internal evaluation criteria, informal political capital within the house, and nonforecast tasks like client service and marketing.
- **Why this matters.** The paper is careful to interpret forecast accuracy and optimism as proxies for broader analyst traits that brokerage houses may care about, not necessarily as the exact variables written into compensation contracts.

3 Models and empirical specifications (all equations + interpretation)

3.1 Absolute forecast accuracy

Let $F_{i,j,t}$ denote analyst i ’s most recent year-end EPS forecast for firm j issued between January 1 and July 1 of year t , and let $A_{j,t}$ be actual earnings per share. The paper defines forecast error as

$$\text{Forecast Error}_{i,j,t} = \frac{|F_{i,j,t} - A_{j,t}|}{P_j}, \quad (1)$$

where P_j is the firm’s stock price.

The analyst’s absolute forecast-accuracy measure is then

$$\text{Absolute Forecast Accuracy}_{i,t} = -\frac{1}{n} \sum_{j \in J} \text{Forecast Error}_{i,j,t}, \quad (2)$$

where the sum is taken over all firms the analyst covers in year t and the prior two years.

Interpretation.

- The scaling by price makes errors comparable across firms.
- The minus sign means *higher values are better*.
- Averaging over three years reduces noise, especially for analysts who cover only a few firms in any given year.

3.2 Relative forecast accuracy

Because analysts cover different firms and some firms are intrinsically harder to forecast, the paper’s main performance measure is *relative* rather than absolute.

For each firm-year, analysts are ranked by forecast error. The best analyst gets rank 1, the worst gets rank $N_{j,t}$, and ties receive midpoint ranks. The score is

$$\text{Score}_{i,j,t} = 100 - \frac{\text{Rank}_{i,j,t} - 1}{N_{j,t} - 1} \times 100, \quad (3)$$

where $N_{j,t}$ is the number of analysts covering firm j in year t .

Interpretation.

- Best analyst on a firm gets a score of **100**.
- Worst analyst gets a score of **0**.
- Median performance is around **50**.

The overall analyst measure is the average of these firm-level scores over year t and the prior two years:

$$\text{Relative Forecast Accuracy}_{i,t} = \frac{1}{n} \sum_{\tau=t-2}^t \sum_{j \in J_{i,\tau}} \text{Score}_{i,j,\tau}. \quad (1)$$

Why this is the preferred measure.

- It compares analysts only against others covering the same firm.
- It strips out firm difficulty.
- It is the measure that strongly predicts career outcomes in the paper.

3.3 Relative forecast optimism

For each firm j covered by analyst i in year t , define

$$I_{i,j,t} = \mathbf{1} \left\{ F_{i,j,t} > \bar{F}_{-i,j,t} \right\}, \quad (2)$$

where $\bar{F}_{-i,j,t}$ is the average forecast of *other* analysts following the same firm.

The analyst's year- t optimism score is the average of these indicators across followed stocks, and the main overall measure is the three-year average:

$$\text{Relative Forecast Optimism}_{i,t} = \frac{1}{n} \sum_{\tau=t-2}^t \sum_{j \in J_{i,\tau}} I_{i,j,\tau}. \quad (3)$$

Interpretation.

- This is not “optimism relative to truth.” It is optimism *relative to the consensus*.
- That is exactly what the paper wants: whether an analyst is the one standing out on the optimistic side and therefore more useful for stock promotion.

Summary statistics.

- Relative accuracy has mean about **50.94**, standard deviation about **8.15**, 10th percentile **40.77**, and 90th percentile **60.61**.
- Relative optimism has mean about **47.58**, standard deviation about **13.51**, 10th percentile **30.77**, and 90th percentile **64.71**.
- The two measures are negatively correlated, about -0.18 . This matters because optimistic analysts tend to be somewhat less accurate, so optimism is *not* just proxying for accuracy.

3.4 Baseline job-separation probit

The paper begins with the conceptual specification

$$\Pr(\text{Job Separation}_{i,t+1} = 1) = \Phi\left(\alpha + \beta_1 \text{Forecast Accuracy Indicator}_{i,t}\right), \quad (4)$$

but quickly moves to a richer model:

$$\begin{aligned} \Pr(\text{Job Separation}_{i,t+1} = 1) = \Phi\Big(&\alpha + \beta_1 \text{Forecast Accuracy Indicator}_{i,t} + \text{Average Coverage Effects}_{i,t} \\ &+ \text{Number of Firms Covered Effects}_{i,t} + \text{Experience Effects}_{i,t} + \text{Year Effects}_{t+1} + \end{aligned} \quad (5)$$

Interpretation.

- The dependent variable is alternately:
 - move down the hierarchy,
 - move up the hierarchy,
 - or other job-separation outcomes.
- The coefficient of interest is the response of future career outcomes to lagged forecast performance.

3.5 Why the controls are necessary

The controls are not cosmetic. They solve concrete problems:

- Analysts covering **few firms** are more likely to land in the tails of the performance distribution by luck.
- Analysts covering **thinly followed firms** are more likely to get extreme relative scores because there are fewer competitors on each stock.
- Experience matters because labor markets naturally involve upward funneling and early-career turnover.
- Brokerage-house fixed effects absorb persistent differences in internal culture, prestige, and labor-market position.

The paper implements:

- average-coverage controls based on the average analyst following of the firms in the analyst’s portfolio,
- dummies for how many firms the analyst covers,
- experience dummies,
- year dummies,
- brokerage-house dummies,
- and brokerage-house clustered standard errors.

3.6 Optimism specification

To test whether optimism matters beyond skill, the paper estimates

$$\begin{aligned} \Pr(\text{Job Separation}_{i,t+1} = 1) = \Phi \Big(& \alpha + \beta_1 \text{Forecast Optimism Indicator}_{i,t} + \text{Relative Accuracy Effects}_{i,t} \\ & + \text{Average Coverage Effects}_{i,t} + \text{Number of Firms Covered Effects}_{i,t} + \text{Experience}_{i,t} \\ & + \text{Year Effects}_{t+1} + \text{Brokerage House Effects}_{i,t} \Big). \end{aligned} \quad (6)$$

Interpretation.

- This asks whether, among analysts with similar measured accuracy, the more optimistic ones have better career outcomes.
- That is the central test of the “brokerages reward stock promotion” hypothesis.

3.7 Interaction models: underwriting, time, and star status

To see whether the payoff to accuracy or optimism varies across analyst environments, the paper estimates interaction models.

For accuracy:

$$\Pr(\text{Job Separation}_{i,t+1} = 1) = \Phi\left(\alpha + \beta_1 \text{Forecast Accuracy Indicator}_{i,t} + \beta_2 \text{Analyst Characteristic}_{i,t} + \beta_3 \text{Forecast Accuracy Indicator}_{i,t} \times \text{Analyst Characteristic}_{i,t} + \text{controls}\right). \quad (7)$$

For optimism:

$$\Pr(\text{Job Separation}_{i,t+1} = 1) = \Phi\left(\alpha + \beta_1 \text{Forecast Optimism Indicator}_{i,t} + \beta_2 \text{Analyst Characteristic}_{i,t} + \beta_3 \text{Forecast Optimism Indicator}_{i,t} \times \text{Analyst Characteristic}_{i,t} + \text{Relative Accuracy}_{i,t}\right). \quad (8)$$

The **Analyst Characteristic** is alternately:

- percent of the analyst's portfolio underwritten by her brokerage house,
- an indicator for the post-1995 period,
- an indicator for All-American status.

Interpretation.

- β_3 tells us whether the payoff to accuracy or optimism is stronger or weaker in the environment of interest.

3.8 Alternative career-concern measure: internal assignments

The paper also studies whether analysts are removed from important stocks inside the house:

$$\Pr(\text{Analyst Stops Covering Stock}_{i,j,k,t} = 1) = \Phi\left(\alpha + \beta_1 \text{Forecast Accuracy Indicator}_{i,t} + \text{Average Coverage Effects}_{i,t} + \text{Number of Firms Covered Effects}_{i,t} + \text{Year Effects}_t + \text{Brokerage Effects}_t\right). \quad (9)$$

Here, a stock is defined as **high-profile** if it is either:

- followed by at least **20 analysts**, or
- has market capitalization above **\$5 billion**.

Interpretation.

- Losing coverage of a marquee stock like Microsoft is an internal career setback even if the analyst stays at the same brokerage house.
- This gives the paper a second, within-house measure of career concerns.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main effects

This is not an instrument-based paper. Identification comes from **predictive variation in lagged forecast behavior** after controlling for many nonperformance confounds.

- Relative forecast accuracy is measured *before* the career outcome.
- The paper compares analysts within the same labor market environment after conditioning on experience, brokerage house, year, and the structure of the analyst's portfolio.
- The relative-score design helps compare analysts who cover different firms by normalizing against others covering the same firm.

4.2 What makes the design credible

- **Future outcome, lagged behavior.** Forecast measures are computed from prior years and related to year- $t + 1$ career outcomes.
- **Rich controls for portfolio composition.** Because extreme measured performance can come mechanically from covering fewer or thinner-followed firms, the paper controls directly for average coverage and number of firms followed.
- **Experience controls.** Upward mobility and turnover vary systematically with analyst age in the profession.
- **Brokerage-house fixed effects.** These absorb time-invariant differences across houses, such as prestige, compensation culture, or screening intensity.
- **Robustness across status measures.** Results do not hinge on one particular ranking of brokerage prestige.
- **Alternative career measure.** The stock-assignment evidence inside houses lines up with the across-house mobility evidence.

4.3 What the paper can and cannot claim

- The paper can claim that career outcomes are systematically related to lagged accuracy and optimism in a way consistent with labor-market incentives.
- It cannot cleanly claim a fully causal effect of one extra unit of optimism holding all else fixed in the experimental sense, because optimism may proxy for unobserved traits like persuasiveness, client relationships, or overall promotional skill.
- The paper also cannot observe compensation directly, so job transitions are proxies for career rewards rather than exact wage effects.

4.4 Main limitations

- **Unobserved tasks.** Analysts do much more than issue earnings forecasts. Client service, stock recommendations, and media presence may also drive job mobility.
- **Selection into brokerage houses.** Better analysts may be hired by better houses early, and unobserved matching may remain even with controls.
- **No explicit-contract evidence.** The paper documents implicit incentives but not the formal pay formulas behind them.
- **Career-outcome proxy.** Moving between houses is informative, but not all career success shows up this way. Some successful analysts stay put and get paid more instead.

4.5 Why the results are still persuasive

- Accuracy predicts favorable moves in exactly the direction a career-concern model would suggest.
- Optimism predicts favorable moves even after conditioning on accuracy.
- The optimism effect strengthens where conflicts of interest should be strongest: affiliated underwriting situations and the late-1990s boom.
- The within-house stock-assignment results point in the same direction.

5 Tables

5.1 Table I and Table II: the brokerage hierarchy is real and stable enough to study

Table I
Characteristics of Brokerage Houses over Time

The entries are descriptive statistics on brokerage houses in the I/B/E/S database between 1983 and 2000. For each year in the sample, we report the total number of such houses and sample statistics on the size (number of analysts employed) of these houses.

Year	Number of Houses	Number of Analysts Working for a Brokerage House			
		Average	25th Percentile	Median	75th Percentile
1983	90	20.71	7	12	25
1984	108	18.26	7	11	25
1985	126	16.56	5	10	22
1986	136	15.02	4	9	16
1987	142	15.07	5	9	17
1988	153	13.46	5	9	16
1989	171	12.67	4	8	15
1990	174	11.91	4	7	15
1991	178	10.76	4	7	14
1992	166	11.83	4	7	15
1993	196	11.41	3	7	15
1994	196	11.57	3	7	14
1995	211	12.48	3	7	15
1996	223	12.66	3	7	15
1997	278	10.70	2	5	12
1998	317	11.04	2	5	11
1999	314	11.54	1	5	12
2000	305	11.61	1	5	12

Table II
Percentage of Analysts Who Work for High-Status Brokerage Houses

The entries are the percentage of all analysts who are categorized as working for high-status brokerage houses. For each year in the sample, we report these percentages for each of four three status measures: the I.I., size, and Carter-Manaster rankings, respectively.

Year	I.I. Ranking	Size Ranking	Carter-Manaster Ranking
1983	22.87	39.13	18.57
1984	23.66	33.44	20.87
1985	21.34	29.76	19.13
1986	24.91	31.94	23.00
1987	20.99	28.12	22.91
1988	26.44	28.69	23.90
1989	26.35	28.11	24.69
1990	21.78	25.07	20.67
1991	23.19	25.03	19.41
1992	20.66	25.68	19.34
1993	24.84	25.97	20.15
1994	24.74	24.88	19.97
1995	22.78	25.20	19.87
1996	22.81	25.92	19.74
1997	21.37	24.70	19.72
1998	24.49	25.76	19.65
1999	21.79	27.73	19.24
2000	24.81	25.86	18.50

Read: Table I gives the evolving size distribution of brokerage houses; Table II shows the fraction of analysts employed by houses deemed high-status under different prestige rankings.

- The number of houses rises sharply over time, from **90** to more than **300**.
- Average house size falls from about **20.7** analysts to about **11.6**.

- Under the main I.I. measure, high-status houses employ about **one-fifth to one-quarter** of analysts each year.

Why it matters.

- The high-status category is selective enough to be meaningful.
- At the same time, it is not so tiny that there are no mobility patterns to study.

5.2 Table III: analysts move a lot, but upward mobility is rare

Read: This table reports the annual distribution of analyst job separations.

- Around **14%** of analysts switch houses each year.
- Among experienced analysts, the unconditional probability of moving *up* is about **2.73%**; moving *down* is about **7.77%**.
- Most moves are not glamorous upward jumps. Many are lateral, especially low-to-low moves.

Why it matters.

- The analyst labor market is active enough that career incentives should matter.
- But because upward moves are rare, it is a demanding outcome to explain.

Table III
Summary Statistics of Analyst Job Separations

Security analysts in the I/B/E/S database in a year are tracked to see if they separate from their employer during the next year. We report the percentage of analysts who experience various types of job separations in a year (averaged over the sample period of 1983 to 2000). These percentages are calculated using the I.I. rankings to determine brokerage house status. In Panel A, the sample includes all analysts. In Panel B, the sample includes analysts with at least three years of experience.

Panel A: Entire I/B/E/S Sample			
	% of Analysts Who Change Houses Each Year: 14.32%	% of Analysts Working for Low-Status House	% of Analysts Working for High-Status House
Analysts Who Work for a Low-Status House	12.15%	2.20%	
Who Move to a High-Status House			
Analysts Who Work for a High-Status House	10.25%		7.02%
Who Move to a Low-Status House			
Analysts Who Work for a High-Status House	5.09%		3.49%
Who Move to Another High-Status House			
Analysts Who Work for a Low-Status House	72.50%	13.13%	
Who Move to Another Low-Status House			
Panel B: Analysts with More Than Three Years of Experience			
	% of Analysts Who Change Houses Each Year: 14.43%	% of Analysts Working for Low-Status House	% of Analysts Working for High-Status House
Analysts Who Work for a Low-Status House	14.53%	2.73%	
Who Move to a High-Status House			
Analysts Who Work for a High-Status House	12.54%		7.77%
Who Move to a Low-Status House			
Analysts Who Work for a High-Status House	6.40%		3.97%
Who Move to Another High-Status House			
Analysts Who Work for a Low-Status House	66.52%	12.52%	
Who Move to Another Low-Status House			

Table IV
Measures of Forecast Performance

Panel A. A Hypothetical Example of a Relative Accuracy Score Calculation
The entries are an example of the forecasts of eight analysts covering a hypothetical firm. The analysts are ranked based on the size of the error of their forecasts, and the relative accuracy score measure of each analyst, described in Section III, is calculated.

Analyst	Forecast Error	Rank	Score
1	0.12	1	100
2	0.25	3	71.4
3	0.25	3	71.4
4	0.25	3	71.4
5	0.38	5	42.9
6	0.67	6.5	21.4
7	0.67	6.5	21.4
8	0.80	8	0

Panel B. Summary Statistics of Analyst Performance Measures
The entries are summary statistics of the two analyst forecast performance measures for analysts with at least three years of experience in the I/B/E/S data set.

	Average (1)	10 th Percentile (2)	Median (3)	90 th Percentile (4)
Analyst's Accuracy Score	50.94 [8.15]	40.77	51.24	60.61
Analyst's Optimism Score	47.58 [13.51]	30.77	47.62	64.71

5.3 Table IV: the two key performance objects

Read: Panel A shows how the relative-accuracy score is constructed. Panel B gives summary statistics for relative accuracy and relative optimism.

- The relative score translates within-firm rank into a **0 to 100** scale.
- Relative accuracy has mean about **51**; relative optimism has mean about **48**.
- The dispersion is meaningful: the 10th-to-90th percentile spread is about **20 points** for accuracy and over **30 points** for optimism.

Why it matters.

- The paper's main empirical content hinges on these two measures.
- The ranking approach is what makes comparisons across different covered firms credible.

5.4 Table V and Table VI: analysts carry their expertise with them, and careers funnel upward

Table V
The Percentage of an Analyst's Portfolio in a Year that Consists of New Stocks

Security analysts in the I/B/E/S database in a year are tracked to see what percentage of stocks they cover in the following year that are not stocks they are covering this year. The entries contain the percentage of an analyst's portfolio the following year that are new stocks. We report these percentages for the entire sample and for analysts who experience various types of separations. These percentages are calculated using the I.I. rankings to determine brokerage house status.

	Percentage of Portfolio that is New
Entire Sample	26.18%
Analysts Who Leave Brokerage Houses	27.59%
Analysts Who Stay with Brokerage Houses	25.93%
	Percentage of Portfolio that is New For Analysts Who Change Houses
Analysts Who Work for a Low-Status House Who Move to a High-Status House	23.73%
Analysts Who Work for a High-Status House Who Move to a Low-Status House	24.50%
Analysts Who Work for a High-Status House Who Move to Another High-Status House	25.74%
Analysts Who Work for a Low-Status House Who Move to Another Low-Status House	29.20%

Table VI

The Percentage of Analysts Who Work for High-Status Brokerage Houses by Experience

Security analysts in the I/B/E/S database who start their career between 1983 and 1999 are partitioned into different samples based upon the number of years they are in the I/B/E/S database. The samples include all analysts who are in the I/B/E/S database a minimum number of years. The entries are the percentage of analysts in these samples that work for high-status brokerage houses by their experience, where high status is measured using the I.I. ranking.

	Minimum Number of Years Analyst Is in Sample									
Years of Experience	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
1	18.32	19.42	19.00	18.79	19.01	19.15	20.10	21.51	22.32	
2	19.77	20.36	20.23	20.15	20.33	20.66	21.41	22.51	23.88	
3		20.89	20.62	20.64	20.91	20.94	21.89	22.79	24.22	
4			22.47	21.93	21.57	21.13	22.49	23.36	23.88	
5				23.35	22.48	22.83	23.56	24.64	25.26	
6					22.48	23.30	23.44	23.93	24.22	
7						22.45	24.04	24.50	25.43	
8							24.52	25.78	25.78	
9								24.93	24.91	
10									26.82	
Number of Analysts	4,907	3,212	2,274	1,623	1,210	1,060	836	702	578	

Read: Table V asks whether movers drastically change coverage portfolios; Table VI tracks the fraction of analysts at high-status houses by experience.

- The new-stock share is roughly similar for movers and stayers, around **26%–28%**.
- The share of analysts at high-status houses rises with experience, from roughly **19%** early in the career to the **mid-20s** later on.

Why it matters.

- Brokerage houses appear to hire analysts for the expertise they built elsewhere.
- Experience is clearly related to career outcomes, so the regressions must control for it.

5.5 Table VII: accuracy predicts who moves up and who moves down

Read: This is the central accuracy table.

- **Move down.** Being in the **bottom 10%** of relative accuracy raises the chance of moving down by about **4.78 percentage points**. Relative to the baseline move-down rate, this is about a **62% increase**.
- **Move up.** Being in the **bottom 10%** of relative accuracy lowers the chance of moving up by about **1.41 percentage points**, roughly a **52% decline** relative to baseline.
- **Top performance matters too.** Being in the **top 10%** of relative accuracy raises the chance of moving up by about **1.12 percentage points**, about a **41% increase**.

- **Relative beats absolute.** When the authors replace relative accuracy with absolute accuracy, predictive power weakens substantially.

Why it matters.

- This is the paper's cleanest evidence that brokerage houses reward forecasting expertise.
- The stronger performance of *relative* rather than *absolute* accuracy also makes economic sense: employers likely care whether an analyst is good *relative to peers*, not merely whether covered firms were easy to forecast.

Table VII
The Effect of Past Accuracy on Job Separations

Security analysts who have at least three prior years of experience are tracked to examine if past forecasting accuracy affects the likelihood that an analyst moves from a high- to a low-status house (move down) or that an analyst moves from a low- to a high-status house (move up). The probit specification is equation (5). In columns (1)–(6), the *Relative Forecast Accuracy* score is used to measure forecasting accuracy. In columns (7)–(10), the *Absolute Forecast Accuracy* score is used to measure forecasting accuracy. Brokerage house status is measured using the I.I. ranking. Standard errors are in parentheses; they are adjusted to account for within-brokerage house correlation of the observations. The entries in the brackets are the marginal probabilities that an analyst with the various accuracy scores experiences a job change compared to other analysts. (*Significant at 10 percent level. **Significant at 5 percent level.)

	Relative Forecast Accuracy						Absolute Forecast Accuracy			
	Moves down			Moves up			Moves down		Moves up	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bottom 10% of Accuracy Scores Indicator	0.3487** (0.1509) [0.0478]	0.5238** (0.2323) [0.0783]	-0.4563** (0.1836) [-0.0411]	-0.5990** (0.1351) [-0.0161]	-0.0428 (0.1776) [-0.0061]	-0.2044 (0.2390) [-0.0085]				
10–25% of Accuracy Scores Indicator		0.2332 (0.1838) [0.0261]		0.1339 (0.1852) [-0.0142]						
25–50% of Accuracy Scores Indicator		0.0966 (0.1981) [0.0110]		-0.1318 (0.1046) [-0.0073]						
50–75% of Accuracy Scores Indicator		0.2471 (0.2190) [0.0000]		-0.1118 (0.0971) [-0.0053]						
75–90% of Accuracy Scores Indicator		0.2526 (0.1716) [0.0319]		-0.0041 (0.1158) [-0.0002]						
Top 10% of Accuracy Scores Indicator		-0.2596 (0.1809) [-0.0246]		0.2053** (0.0864) [0.0112]			-0.0944 (0.1890) [-0.0100]		0.1536 (0.1876) [0.0086]	
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Experience Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Brokerage House Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of Firms Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Average Coverage Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Log Likelihood	-411.44	-412.86	-410.03	-403.68	-403.54	-402.94	-410.07	-412.87	-404.84	-404.81
Observations	1,866	1,866	1,866	6,143	6,143	6,143	1,866	1,866	6,143	6,143

Table VIII
The Effect of Past Optimism on Analyst Job Separations

Security analysts who have at least three prior years of experience are tracked to examine if past forecasting optimism affects the likelihood that an analyst moves from a high- to a low-status house (move down) or that an analyst moves from a low- to a high-status house (move up). The probit specification is equation (6). The measure of optimism is the *Relative Forecast Optimism* score. Brokerage house status is measured using the I.I. ranking. Standard errors are in parentheses; they are adjusted to account for within-brokerage house correlation of the observations. The entries in the brackets are the marginal probabilities that an analyst with the various optimism scores experiences a job change compared to other analysts. (*Significant at 10 percent level. **Significant at 5 percent level.)

	Moves down		Moves up	
	(1)	(2)	(3)	(4)
Top 10% of Relative Forecast Optimism Scores Indicator	-0.4029** (0.1420) [-0.0323]		0.3659** (0.1801) [0.0253]	
Bottom 10% of Relative Forecast Optimism Scores Indicator		0.0400 (0.1624) [0.0044]		-0.0401 (0.2252) [-0.0029]
Bottom 10% of Accuracy Scores Indicator	0.5327** (0.2295) [0.0807]	0.5257** (0.2340) [0.0797]	-0.6420** (0.1638) [-0.0166]	-0.5858** (0.1591) [-0.0159]
10–25% of Accuracy Scores Indicator	0.2367 (0.1838) [0.0295]	0.2347 (0.1818) [0.0294]	-0.4703** (0.1402) [-0.0145]	-0.4380** (0.1416) [-0.0139]
25–50% of Accuracy Scores Indicator	0.1001 (0.1987) [0.0114]	0.0983 (0.1915) [0.0112]	-0.1912* (0.1054) [-0.0074]	-0.1737* (0.1043) [-0.0069]
50–75% of Accuracy Scores Indicator	0.2455 (0.2176) [0.0297]	0.2486 (0.2151) [0.0303]	-0.1326 (0.0975) [-0.0053]	-0.1238 (0.0999) [-0.0050]
75–90% of Accuracy Scores Indicator	0.2489 (0.1679) [0.0312]	0.2535 (0.1713) [0.0320]	-0.0106 (0.1152) [-0.0004]	-0.0043 (0.1155) [-0.0002]
Year Effects	Yes	Yes	Yes	Yes
Experience Effects	Yes	Yes	Yes	Yes
Brokerage House Effects	Yes	Yes	Yes	Yes
Number of Firms Covered Effects	Yes	Yes	Yes	Yes
Average Coverage Effects	Yes	Yes	Yes	Yes
Log Likelihood	-409.08	-410.03	-401.27	-402.87
Observations	1,866	1,866	6,143	6,143

5.6 Table VIII: optimism is rewarded even after controlling for accuracy

Read: This table adds optimism to the story.

- Being in the **top 10%** of relative optimism lowers the chance of moving down by about **3.23 percentage points**.
- It raises the chance of moving up by about **2.53 percentage points**, roughly a **90% increase** relative to the baseline move-up rate.
- Extreme pessimism is not robustly punished; the optimism effect is asymmetric.

Why it matters.

- The labor market is not rewarding only expertise.
- It is rewarding analysts who are unusually upbeat relative to their peers, consistent with the idea that brokerages value promotion.

5.7 Table IX: underwriting relationships shift incentives toward optimism

Read: This table interacts performance with the fraction of the analyst's portfolio involving house-underwritten stocks.

- If an analyst covers no underwritten stocks, bad accuracy substantially raises the chance of moving down.
- As the share of underwritten stocks rises, that accuracy penalty becomes smaller.
- Conversely, optimism becomes more valuable when the analyst covers more affiliated stocks.
- A one-standard-deviation increase in affiliated-underwriting exposure (about **5%** of the portfolio) noticeably weakens the penalty from bad accuracy and strengthens the reward to optimism.

Why it matters.

- This is the most direct conflict-of-interest evidence in the paper.
- It shows that the reward system changes exactly where investment-banking pressures should be strongest.

Table IX
The Effect of Accuracy and Optimism on Analyst Job Separations by Underwriter Status

Security analysts who have at least three prior years of experience are tracked to examine if the effects of relative forecasting accuracy and optimism on the likelihood that an analyst moves from a high- to a low-status house (move down) depend on underwriting relationships. The probit specification is equation (7). *Percent Underwriting* is the fraction of an analyst's portfolio of stocks that have underwriting relationships with her brokerage house. Brokerage house status is measured using the I.I. ranking. Standard errors are in parentheses; they are adjusted to account for within-brokerage house correlation of the observations. The entries in the brackets are the marginal probabilities that an analyst with the various accuracy and optimism scores and underwriting relationships experiences a move down compared to other analysts. (*Significant at 10 percent level. **Significant at 5 percent level.)

	Moves down	Moves down
	(1)	(2)
Bottom 10% of <i>Relative Forecast Accuracy</i> Scores Indicator	0.4287** (0.1591) [0.0615]	
<i>Percent Underwriting</i>	0.7172 (1.221) [0.0761]	1.621 (1.240) [0.1539]
Bottom 10% of <i>Relative Forecast Accuracy</i> Scores Indicator $\times$ <i>Percent Underwriting</i>	-4.232* (2.361) [-0.4430]	
Top 10% of <i>Relative Forecast Optimism</i> Scores Indicator		-0.2091 (0.2073) [-0.0172]
Top 10% of <i>Relative Forecast Optimism</i> Scores Indicator $\times$ <i>Percent Underwriting</i>		-6.527* (3.948) [-0.6194]
Relative Forecast Accuracy Effects	No	
Year Effects	Yes	Yes
Experience Effects	Yes	Yes
Brokerage House Effects	Yes	Yes
Number of Firms Covered Effects	Yes	Yes
Average Coverage Effects	Yes	Yes
Log Likelihood	-403.63	-402.64
Observations	1,866	1,866

Table X
The Effect of Accuracy and Optimism on Analyst Job Separations by Subperiods

Security analysts who have at least three prior years of experience are tracked to examine if the effects of relative forecasting accuracy and optimism on the likelihood that an analyst moves from a high- to a low-status house or from a low- to a high-status house differ between 1996 to 2000 and early periods. The probit specification is equation (7). *After 1995 Indicator* equals one for analysts' forecasts issued after 1995. Brokerage house status is measured using the I.I. ranking. Standard errors are in parentheses; they are adjusted to account for within-brokerage house correlation of the observations. The entries in the brackets are the marginal probabilities that an analyst with the various accuracy and optimism scores issuing forecasts in different periods experiences a move down compared to other analysts. (*Significant at 10 percent level. **Significant at 5 percent level.)

	Moves down		Moves up	
	(1)	(2)	(3)	(4)
Bottom 10% of <i>Relative Forecast Accuracy</i> Scores Indicator	0.5336** (0.1767) [0.0813]		-0.5959** (0.1685) [-0.0210]	
Bottom 10% of <i>Relative Forecast Accuracy</i> Indicator $\times$ <i>After 1995 Indicator</i>	-0.4462* (0.2529) [-0.0353]		0.3578 (0.3129) [0.0277]	
Top 10% of <i>Relative Forecast Optimism</i> Scores Indicator		-0.2395 (0.2075) [-0.0196]		0.3307 (0.2373) [0.0224]
Top 10% of <i>Relative Forecast Optimism</i> Scores Indicator $\times$ <i>After 1995 Indicator</i>		-0.4157 (0.2852) [-0.0290]		0.1702 (0.3764) [0.0113]
Relative Forecast Accuracy Effects	No	Yes	No	Yes
Year Effects	Yes	Yes	Yes	Yes
Experience Effects	Yes	Yes	Yes	Yes
Brokerage House Effects	Yes	Yes	Yes	Yes
Number of Firms Covered Effects	Yes	Yes	Yes	Yes
Average Coverage Effects	Yes	Yes	Yes	Yes
Log Likelihood	-410.32	-403.66	-402.35	-401.05
Observations	1,866	1,866	6,143	6,143

5.8 Table X: the late-1990s boom weakened the reward to accuracy

Read: This table compares pre-1996 and post-1995 periods.

- In **1986–1995**, being in the bottom 10% of accuracy raises the chance of moving down by about **8.1 percentage points**.
- After 1995, that effect falls to roughly **4.6 percentage points**.

- The reward to optimism becomes economically stronger after 1995 as well, though the statistical precision is weaker.

Why it matters.

- The analyst labor market appears to have shifted during the stock-market mania.
- Wall Street seems to have become less disciplined about objective research in that period.

5.9 Table XI: All-Americans are partly insulated from poor accuracy

Read: This table asks whether star analysts face different incentives.

- For non-All-Americans, poor accuracy meaningfully raises the chance of moving down.
- For All-Americans, that penalty is much smaller.
- The optimism reward does not differ much by All-American status.

Why it matters.

- Reputation and star status provide some insulation from the labor-market consequences of bad forecasting.
- But the broader optimism reward does not appear to be driven only by stars.

Table XI
The Effect of Accuracy and Optimism on Analyst Job Separations by All-American Status

Security analysts who have at least three prior years of experience are tracked to examine if the effects of relative forecasting accuracy and optimism on the likelihood that an analyst moves from a high- to a low-status house (move down) depend on whether an analyst is an All-American. The probit specification in columns (1) and (2) are equations (7) and (8), respectively. *All-American Indicator* equals one if an analyst became an All-American the previous year. Brokerage house status is measured using the I.I. ranking. Standard errors are in parentheses; they are adjusted to account for within-brokerage house correlation of the observations. The entries in the brackets are the marginal probabilities that an analyst with the various accuracy and optimism scores and All-American status experiences a move down compared to other analysts. (*Significant at 10 percent level. **Significant at 5 percent level)

	Moves down (1)	Moves down (2)
Bottom 10% of Relative Forecast Accuracy Scores Indicator	0.3716** (0.1569) [0.0516]	-0.3872** (0.1521) [-0.0283]
<i>All-American Indicator</i>	-0.0234 (0.1220) [-0.0025]	
Bottom 10% of Relative Forecast Accuracy Scores Indicator $\times$ <i>All-American Indicator</i>	-0.4848** (0.2218) [-0.0362]	
Top 10% of Relative Forecast Optimism Scores Indicator		-0.4144** (0.1501) [-0.0297]
Top 10% of Relative Forecast Optimism Scores Indicator $\times$ <i>All-American Indicator</i>		0.1441 (0.5236) [0.0154]
Relative Forecast Accuracy Effects	No	Yes
Year Effects	Yes	Yes
Experience Effects	Yes	Yes
Brokerage House Effects	Yes	Yes
Number of Firms Covered Effects	Yes	Yes
Average Coverage Effects	Yes	Yes
Log Likelihood	-411.13	-403.76
Observations	1,866	1,866

Table XII
The Effect of Accuracy on Whether a Brokerage House Removes an Analyst from Following a Firm

Brokerage houses that cover a firm for more than one year are tracked to see whether the accuracy of the analyst following the firm influences whether the brokerage house replaces that analyst with another analyst. The probit specification is equation (9). The sample includes all analysts that continue to work for the same brokerage house and cover the same industry as they did the previous year. Regression in (1) measures the effect of poor relative accuracy on whether an analyst is removed from following the firm. Regression in (2) measures the same effect only for analysts that follow firms that are covered by at least 20 other analysts. Regression in (3) measures the same effect for analysts covering firms worth more than \$5 billion. Regressions in (4)-(6) are identical to (1)-(3) except that the effect of good relative accuracy is measured. The entries in the brackets are the marginal probabilities that an analyst is removed from following a firm. Standard errors are in parentheses; they are adjusted to account for within-brokerage house correlation of the observations. (*Significant at 10 percent level. **Significant at 5 percent level)

	Poor relative accuracy			Good relative accuracy		
	All stocks (1)	High coverage (2)	High value (3)	All stocks (4)	High coverage (5)	High value (6)
Indicator for Poor Past Performance (Bottom 10% of Distribution)	0.0321 (0.0434) [0.0042]	0.1732** (0.0849) [0.0245]	0.1568* (0.0826) [0.0213]			
Indicator for Good Past Performance (Top 10% of Distribution)				0.0003 (0.0416) [0.0000]	-0.0518 (0.0830) [-0.0064]	-0.0239 (0.0822) [-0.0029]
Year Effects	Yes	Yes	Yes	Yes	Yes	Yes
Experience Effects	Yes	Yes	Yes	Yes	Yes	Yes
Brokerage House Effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of Firms Covered Effects	Yes	Yes	Yes	Yes	Yes	Yes
Average Coverage Effects	Yes	Yes	Yes	Yes	Yes	Yes
Log Likelihood	-24,291.37	-6,607.70	-7,093.31	-24,292.32	-6,614.45	-7,099.86
Observations	110,077	34,028	26,439	110,077	34,028	26,439

5.10 Table XII: internal labor markets reinforce the same message

Read: This table examines whether analysts are removed from important stocks inside their firms.

- For **all stocks**, poor performance has essentially no effect on whether an analyst stops covering the stock.
- For **high-profile stocks**—those with analyst coverage of at least 20 or value above **\$5 billion**—poor performance raises the chance of removal by about **two percentage points or more**.
- Good performance works in the opposite direction, though less precisely.
- Optimism has similar but somewhat smaller and noisier effects.

Why it matters.

- The same incentive pattern shows up inside houses, not only across houses.
- That makes the paper’s interpretation more convincing: firms track desirable analysts into marquee assignments.

6 Short summary

- **Method.** Build relative measures of analyst accuracy and optimism from I/B/E/S forecast histories, then relate them to future job moves across a brokerage-house prestige hierarchy.
- **Main result 1.** Accurate analysts are more likely to move up and less likely to move down.
- **Main result 2.** Controlling for accuracy, relatively optimistic analysts are also rewarded.
- **Main result 3.** These optimism rewards are stronger, and accuracy rewards weaker, when conflicts of interest are more salient: affiliated underwriting, the late 1990s, and high-profile internal coverage assignments.
- **Big lesson.** Analyst labor markets reward both information production and stock promotion.

7 Conclusion

Core conclusion. The paper shows that analyst career concerns are not one-dimensional. Brokerage houses reward forecast expertise, but they also reward optimism relative to the consensus. The balance between those two rewards shifts toward optimism precisely where conflicts of interest should be strongest.

Economic intuition. A brokerage wants analysts who are credible enough to be listened to, but also helpful enough to generate underwriting business and trading commissions. That combination implies rewarding analysts who are competent *and* marketable. Accuracy builds reputation; optimism helps promote stocks.

Takeaway. This is an early and important paper on the incentive problem inside financial intermediation. It suggests that debates about analyst bias should not focus only on explicit compensation or formal underwriting ties. Implicit labor-market incentives can also push analysts toward optimism.